BioCONTINUE - Short Cut

BIOTRONIK
excellence for life

BIOtronik study to assess the **CONTINU**ation of **E**xisting risk of ventricular arrhythmias after CRT-D replacement for patients with primary prevention indication

Version 1-0/Oct 20, 2014

BACKGROUND

Cardiac resynchronization therapy (CRT) has been shown to reduce heart failure (HF), hospitalizations and death in patients with left ventricular ejection fraction (LVEF) <35% and wide QRS. CRT provides electromechanical resynchronization and improves LV systolic function. The induced LV reverse remodeling or near normalization in LVEF to ≥45% is associated with a significant reduction in the risk of subsequent life-threatening **ventricular tachyarrhythmias (VTA)**. And at the time of replacement, the need for defibrillator back-up after an event-free first CRT-D service-life for patients with improved LVEF is a controversy question.

80% of ICD patients implanted for **primary prevention** do not experience VTA during the lifetime of their first device.

So, regarding patients implanted with a CRT-D for primary prevention at the time of first implantation, the question is will they experience VTA after their device replacement by another CRT-D.

STUDY OBJECTIVE

The objective of this study is to describe, in a population of patients who had a primary prevention ICD indication at first implantation of a CRT-D, the relevance defibrillator back-up after the replacement of the first CRT-D. Furthermore, the association between the baseline characteristics of the CRT-D population after replacement and the risk of subsequent VTA will be explored after a minimum of two years FU.

Primary objective:
Assess the rate of patients with **at least one sustained VTA** (sustained ventricular tachycardia (VT) or fibrillation (VF) detected by the CRT-D or a conventional surface ECG, holter ECG…).

Secondary objective: subgroup analysis
Rate of patients with at least one VTA for the 4 following subgroups:

- Grp R[1-4]: LVEF ≥40% / <40% at the time of replacement and with / without previous VTA during the lifetime of their first CRT-D device.

Tertiary objectives:

- Rate of patients with ventricular tachycardia (VT) or ventricular fibrillation (VF) events, with **appropriate therapies**
- Rate of patients with **inappropriate therapies**
- Rate of patients with inappropriate diagnoses
- Type and etiology of inappropriate therapies
- Sustained VTA non diagnosed by the ICD, documented by a conventional ECG…
- Baseline factor predictors of ventricular events (age, sex, comorbidities, aetiology…)
- SAE related to ventricular events
- SAE related to inappropriate therapies

SYSTEMS USED

- CRT-D: Ilesto/Iforia HF-T or previous/next version
- Programmer: ICS 3000/Renamic
- CardioMessenger IIs (highly recommended)
- Home-Monitoring Website (highly recommended) *www.biotronik-homemonitoring.com*
- Data reporting on electronic CRF *www.imednet.com/index.php*

STUDY DESIGN

- **Prospective**
- Non-interventional
- Multicenter
- International

INCLUSION CRITERIA

- Patient with **primary indication** for defibrillation at the time of first CRT-D -implantation
- Patient implanted as first **replacement by a CRT-D device (CRT-D → CRT-D)**
- Lifetime of the previous CRT-D device > 3 years
- Patient willing and able to comply with the protocol and who has provided written informed consent
- Patient whose medical situation is stable.

EXCLUSION CRITERIA

- Patient with an ICD lead under advisory (e.g. Fidelis lead)
- Right or left ventricular leads exchange during CRT-D replacement
- Non functional atrial or right/left ventricular leads
- Life expectancy < 1 year
- Age < 18 years
- Pregnant woman or woman who plan to become pregnant during the trial
- Participation in another clinical study

PATIENTS / CENTERS

- Sample size: **277 patients**
- Approx 8 countries, 30-40 centers
- Beginning: Q1 2015
- End of enrolment (24 months): Q1 2017
- Follow Up : between 2 – 4 years
- End of study: ≈Q1 2019 for all the patients (date of last patient enrolled + 2 years follow-up)

All rights reserved by BIOTRONIK. Copying, even in extracts, is only allowed by license of BIOTRONIK.

BioCONTINUE - Short Cut

BIOTRONIK
excellence for life

BIOtronik study to assess the **CONTINU**ation of **E**xisting risk of ventricular arrhythmias after CRT-D replacement for patients with primary prevention indication

Version 1-0/Oct 20, 2014

STUDY DESIGN

First CRT-D replacement
by another CRT-D device

↓

Enrollment / Hospitalization discharge

↓

IEGM/ECG analysis during in office follow-up or by Home-Monitoring system

↓

Study termination ≈Q1 2019 for all patients

RECOMMENDED PROGRAMMING

The programming of the device is left at the physician's discretion. The programming below is a recommendation in case the physician wishes to program the ICD functions according to the MADIT-RIT study.*

Tachycardia parameters with high-rate therapy and prolonged detection criterion:

Zone 1:
171 bpm, Monitor only

Zone 2:
≥200 bpm, 18/24 for VF detection
ATP one shot - Shock

*Source: Moss A et al. NEJM 2012 (MADIT RIT)

METHODOLOGY

This study is qualified as observational i.e. standard care is provided to the patient. All follow-up visits are at the physician's own discretion and may be performed on patients' or physicians' demand.

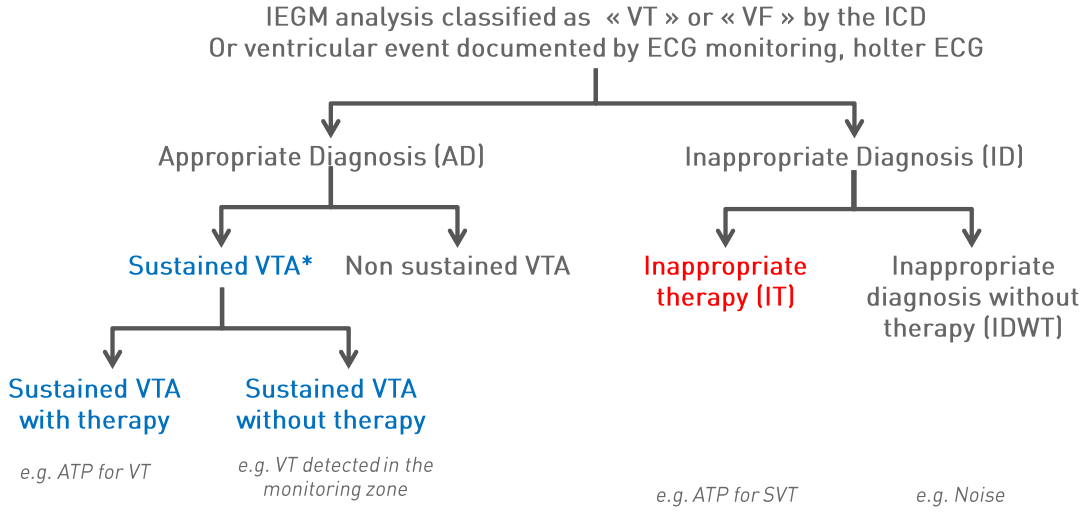
Device replacement
- Medical devices used
- Device programming
- Initial device indication for implant

Enrollment & baseline evaluation
- Inclusion / Exclusion Criteria
- Patient informed consent
- Demographic patient data
- General Health Status at re-implantation
- Documented arrhythmia history
- LVEF (at first implantation and replacement)
- Cardiac disease at implantation
- Current extra-cardiovascular morbidity

Analysis of ventricular events
- Ventricular tachycardia without therapy
- Ventricular tachycardia with therapy
- Ventricular fibrillation without therapy
- Ventricular fibrillation with therapy
- Inappropriate diagnosis without therapy
- Inappropriate diagnosis with therapy

Study termination
- Regular / premature (reason)

FLOW CHART (IEGM/ECG ANALYSIS)



*VTA > 30 sec or interrupted by ICD treatment (*Primary endpoint*)

RESPONSABILITIES

Coordinating investigator
Dr GRAS Daniel
Nouvelles Cliniques Nantaises
44 277 Nantes - FRANCE

Project manager
Sophie Fauquembergue
Biotronik France
+33.(0)1.46.75.39.42

Sponsor
BIOTRONIK SE & Co. KG
Woermannkehre 1
D-12359 Berlin

All rights reserved by BIOTRONIK. Copying, even in extracts, is only allowed by license of BIOTRONIK.